

β Derivation: Complete Status

Formula Exact · $f_{II} = 1/n$ from SO(3) · Born Cascade Correlation
Identified

Gap reduced from 41% to 16%. Remaining gap precisely named.

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Abstract

This document records the complete status of the β derivation after the current session.

Three major results: (1) The β formula is verified exact from per-event simulation data (residual = 0.000 across 2016 events). (2) H_{uniform} and H_{peaked} are fully derived from (ε_0, n) via the digamma formula. (3) The hypothesis $f_{II} = 1/n$ (from Ψ -mode symmetry via SO(3)) gives $H_{\text{after}} = 1.836$, reducing the β gap from 41% to 16%.

The remaining 16% gap is precisely identified as the **Born cascade correlation**: a many-body effect where recently-Born neighbours produce structured $\bar{P}$, reducing H_{after} below the $f_{II} = 1/n$ prediction. The enhancement factor is $e = f_{II}/(1/n) = 1.174$.

Deriving e requires tracking the P_i autocorrelation between R-events, which is a new, well-posed open problem.

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1 Formula Verification: Exact

β Formula Verified Exact (Residual = 0)

Per-event decomposition from V20.3, 2016 events:

$$\begin{aligned}\beta &= \frac{1}{\kappa} \mathbb{E}[F_i(H_{\text{after},i} \ln(m_i + \kappa) - H_{\text{before},i} \ln m_i)] \\ &= \underbrace{2.2548}_{\text{main term}} + \underbrace{(-0.0395 + 0.0418)}_{\text{Cov terms}} = 2.2571\end{aligned}$$

$\beta_{\text{actual}} = 2.2571$. **Residual = 0.0000.**

The covariance correction is negligible (+0.002). The formula is structurally exact. All gap is in H_{after} .

2 Derived: H_{uniform} and H_{peaked}

Digamma Formula from (ε_0, n)

Born resampling: $P_i \sim \text{Dir}(\alpha)$, $\alpha_k = (\bar{P}_k + \varepsilon_0)/\varepsilon_0$.

Uniform neighbours (Phase I):

$$\alpha_k^{(u)} = \frac{1}{n\varepsilon_0} + 1 = 17.55, \quad H_{\text{uniform}} = \psi(n\alpha^{(u)} + 1) - \psi(\alpha^{(u)} + 1) = 2.055$$

Peaked neighbours (Phase II):

$$\alpha_{\text{peak}} = \frac{1}{\varepsilon_0} = 132.5, \quad \alpha_{\text{bg}} = \frac{1}{n} + 1 = 1.125, \quad H_{\text{peaked}} = 0.304$$

Both derive from (ε_0, n) only. No free parameters. ε_0 from EIC/SINDy; n from SO(3).

3 Hypothesis: $f_{II} = 1/n$ from Ψ -Mode Symmetry

$f_{II} = 1/n$: Derived from $\text{SO}(3) \rightarrow n$

V20.3 simulation produces exactly $n = 8$ distinct Ψ -modes (confirmed: 8 histogram peaks in final Ψ distribution).

Argument:

1. n -modes; by symmetry, each holds fraction $1/n$ of nodes.
2. Nodes in the same -mode as the firing node contribute peaked $\bar{P}$.
3. In a well-mixed graph ($k = 20 \gg 1$), same-mode fraction = $1/n$.

$$f_{II} = \frac{1}{n} = \frac{1}{8} = 0.125$$

$$H_{\text{after}}^{(1/n)} = \frac{7}{8} \cdot 2.055 + \frac{1}{8} \cdot 0.304 = 1.836$$

$$\beta^{(1/n)} = 2.617 \quad \text{vs} \quad \beta_{\text{sim}} = 2.257 \quad [\text{gap: } 16\%]$$

Gap reduced from **41% to 16%** by this hypothesis.

4 Remaining Gap: Born Cascade Correlation

$e = 1.174$: Born Autocorrelation Between R-Events

Measured: $f_{II} = 0.147$, predicted by $1/n$: 0.125. Enhancement: $e = f_{II}/(1/n) = 1.174$.

Source identified: Born cascade correlation.

When node i fires at time t , its P_i is set by $\bar{P}_i$. At time $t + \tau$ when neighbour j fires, $\bar{P}_j$ includes the updated $P_i(t)$. If $\tau < \tau_{\text{Born}} = T_{\text{avg}}/k = 1.77$ sweeps, this correlation has not yet decayed. The structured $P_i(t)$ contributes to a non-uniform $\bar{P}_j$, reducing $H_{\text{after},j}$ below H_{uniform} .

Evidence from simulation:

	H_{after}	f_{II} equiv.
Early events (sweeps 0–50, high R-rate)	1.735	0.183
Late events (sweeps 50–100, low R-rate)	1.861	0.110
Overall	1.798	0.147
$f_{II} = 1/n$ prediction	1.836	0.125

The early/late asymmetry confirms the cascade correlation mechanism: high R-rate sweeps produce more structured $\bar{P}$ (stronger Born correlation), reducing H_{after} below the symmetry prediction.

Deriving e : e depends on the P_i autocorrelation function: $\mathcal{C}(\tau) = \text{Corr}(P_i(t), P_i(t + \tau))$. This decays with timescale $\tau_{\text{Born}} = T_{\text{avg}}/k$. Deriving $\mathcal{C}(\tau)$ from the δ -chain requires solving the Born process autocorrelation equation:

$$\frac{d\mathcal{C}}{d\tau} = -\frac{\mathcal{C}}{\tau_{\text{Born}}} + f(\varepsilon_0, n, \mathcal{C})$$

This is a new, well-posed open problem.

5 Complete Status

Component	Status	Value / Source
ε_0	Derived	EIC/SINDy
$n = 8$	Derived	SO(3) #16
$H_{\text{uniform}} = 2.055$	Derived	$\psi(\varepsilon_0, n)$
$H_{\text{peaked}} = 0.304$	Derived	$\psi(\varepsilon_0, n)$
Formula $\beta = \mathbb{E}[F \cdot \Delta]/\kappa$	Exact	Residual = 0
n -modes confirmed	Observed	V20.3
$f_{II} = 1/n$ hypothesis	Hypothesis	-mode symmetry
H_{after} at $f_{II} = 1/n$	Hypothesis	1.836 vs sim 1.798
Enhancement $e = 1.174$	Identified	Born cascade
$\mathcal{C}(\tau)$: Born autocorrelation	Open	New problem
β gap	41% $\rightarrow$ 16%	After $f_{II} = 1/n$

6 Path to Closure

The derivation hierarchy is now fully mapped:

$$\delta \xrightarrow{\text{EIC}} \varepsilon_0 \xrightarrow{\text{SO}(3)} n \xrightarrow{\psi} H_u, H_p \xrightarrow{\Psi\text{-modes}} f_{II} \approx 1/n \xrightarrow{\mathcal{C}(\tau)} f_{II} \text{ exact} \rightarrow H_{\text{after}} \rightarrow \beta$$

The one missing link: derive $\mathcal{C}(\tau)$ — the Born process autocorrelation — from the δ -chain. This gives e and closes the 16% gap.

The formula is exact.

H_{uniform} and H_{peaked} are derived.

$f_{II} = 1/n$ from -mode symmetry reduces the gap to 16%.

The remaining gap has a name: Born cascade correlation.

*Enhancement $e = 1.174$ comes from structured $\bar{P}$
in high- R -rate sweeps.*

Deriving $\mathcal{C}(\tau)$ from δ closes β .